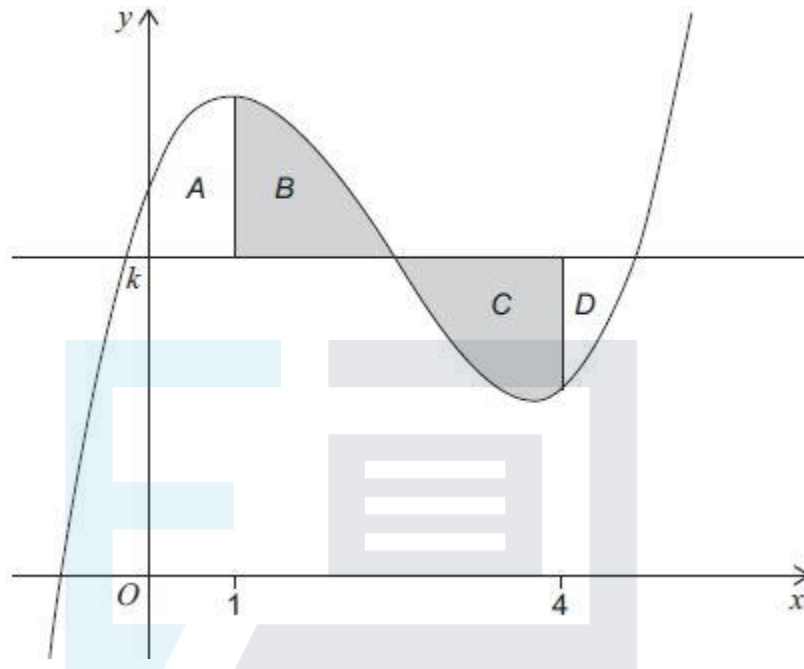


Q1.

Four finite regions A , B , C and D are enclosed by the curve with equation

$$y = x^3 - 7x^2 + 11x + 6$$

and the lines $y = k$, $x = 1$ and $x = 4$, as shown in the diagram below.



The areas of B and C are equal.

Find the value of k .

(Total 3 marks)

Q2.

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Find the mean value of $3x^2$ over the interval $1 \leq x \leq 3$

Circle your answer.

$$8\frac{2}{3}$$

$$10$$

$$13$$

$$26$$

(Total 1 mark)

Q3.

The function $f(x) = \cosh(ix)$ is defined over the domain $\{x \in \mathbb{R} : -a\pi \leq x \leq a\pi\}$, where a is a positive integer.

By considering the graph of $y = [f(x)]^n$, find the mean value of $[f(x)]^n$, when n is an odd positive integer.

Fully justify your answer.

(Total 3 marks)



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Examiner reports

Q1.

Only a minority of students realised that this question was asking for the mean value. A common approach was to translate the graph k units downwards and then solve

$$\int_1^4 f(x) = 0$$

. It was not uncommon for students to assume that the intersection with $y = k$ occurred at $x = 2.5$. However, it was possible to use the unknown intersection point and

$$\text{solve } \int_p^4 y \, dx = -\int_1^p y \, dx$$

. Unfortunately, many students who attempted this method struggled with the subsequent algebra.



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